

Zadání 1. úkolu do předmětu IZU

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Pomocí metody A* najdete nejkratší cestu v mapě složené z pravidelných buněk, kde cena přechodu mezi dvěma stavy (buněkami) je dána číslem, uvedeným v Tabulce 1 (a je stejná pro všechny přechody ze sousedních míst do příslušné buňky). Nepřekročitelné buňky mají hodnotu "Z" (jako "zeď"). Po každém kroku vypište nové hodnoty seznamů Open a Closed. Do pomocné tabulky s ohodnocením uzlů zapisujte aktuálně zkoumaný uzel, cenu cesty do aktuálního uzlu „g“, heuristiku „h“ a celkovou cenu cesty „f“. Heuristiku počítejte jako přímou vzdálenost středů dvou buněk, kde velikost strany jedné buňky je rovna jedné. Uzly generujte v pořadí zleva doprava a shora dolů, uvažujte 8-okolí buňky (tzn. operátory $\nwarrow, \uparrow, \nearrow, \leftarrow, \rightarrow, \swarrow, \downarrow, \searrow$). Výslednou cestu zapište do tabulky Výsledná cesta. Uzel se skládá ze souřadnic, z ohodnocení f a souřadnic uzlu, ze kterého byl vygenerován nebo z operátoru, který byl použit (aby bylo možné nalézt cestu od startu k cíli).

Uzly zapisujte: ([sloupec, řádek], celkové ohodnocení f, [souřadnice otcovského uzlu nebo operátor])

Start: ([3, 5], 5.0, [null])

Cíl: ([6, 1], X, [?, ?])

Výsledná cesta:

[3, 5]	[4, 5]	[5, 4]	[4, 3]	[5, 2]
[6, 1]				

y/x	0	1	2	3	4	5	6	7	8	9
0	8	7	6	9	3	Z	Z	Z	9	8
1	8	7	5	8	3	5	2	Z	9	6
2	8	5	7	8	3	4	6	Z	9	7
3	Z	Z	Z	Z	3	Z	Z	Z	Z	Z
4	9	8	Z	9	9	3	8	Z	9	9
5	7	7	Z	2	4	3	9	Z	9	9
6	7	8	Z	8	7	3	9	Z	9	9
7	8	7	7	8	9	3	9	9	9	9
8	9	9	8	9	7	3	6	7	9	9
9	8	7	6	9	3	9	9	9	9	8

Tabulka 1: Mapa přechodů. Např. cena přechodu do cílové buňky je rovna 2 pro všechny buňky s cílovou buňkou sousedící.

Pomocná tabulka:

	Uzel	g	h	f		Uzel	g	h	f
01.	[3, 5]	0	5.0	5.0	16.	[6, 7]	16	6.0	22.0
02.	[3, 4]	9	4.24	13.24	17.	[3, 7]	15	6.71	21.71
03.	[4, 4]	9	3.61	12.61	18.	[3, 2]	18	3.16	21.16
04.	[4, 5]	4	4.47	8.47	19.	[4, 2]	13	2.24	15.24
05.	[3, 6]	8	5.83	13.83	20.	[5, 2]	14	1.41	15.41
06.	[4, 6]	7	5.39	12.39	21.	[2, 7]	15	7.21	22.21
07.	[5, 4]	7	3.16	10.16	22.	[3, 1]	21	3.0	24.0
08.	[5, 5]	7	4.12	11.12	23.	[4, 1]	16	2.0	18.0
09.	[5, 6]	7	5.10	12.10	24.	[5, 1]	18	1.0	19.0
10.	[4, 3]	10	2.83	12.83	25.	[6, 1]	16	0.0	16.0
11.	[6, 4]	15	3.0	18.0	26.	[6, 2]	20	1.0	21.0
12.	[6, 5]	16	4.0	20.0	27.				
13.	[6, 6]	16	5.0	21.0	28.				
14.	[4, 7]	16	6.33	22.33	29.				
15.	[5, 7]	10	6.08	16.08	30.				

1. iterace

Open:

[3, 5], 5.0, NULL				

Closed:

2. iterace

Open:

[3, 4], 13.24, [3, 5]	[4, 4], 12.61, [3, 5]	[4, 5], 8.47, [3, 5]	[3, 6], 13.83, [3, 5]	[4, 6], 12.39, [3, 5]

Closed:

[3, 5], 5.0, NULL				

3. iterace

Open:

[3, 4], 13.24, [3, 5]	[4, 4], 12.61, [3, 5]	[3, 6], 13.83, [3, 5]	[4, 6], 12.39, [3, 5]	[5, 4], 10.16, [4, 5]
[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]			

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]			

4. iterace

Open:

[3, 4], 13.24, [3, 5]	[4, 4], 12.61, [3, 5]	[3, 6], 13.83, [3, 5]	[4, 6], 12.39, [3, 5]	[5, 5], 11.12, [4, 5]
[5, 6], 12.10, [4, 5]	[4, 3], 12.83, [5, 4]	[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]		

5. iterace

Open:

[3, 4], 13.24, [3, 5]	[4, 4], 12.61, [3, 5]	[3, 6], 13.83, [3, 5]	[4, 6], 12.39, [3, 5]	[5, 6], 12.10, [4, 5]
[4, 3], 12.83, [5, 4]	[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	

6. iterace

Open:

[3, 4], 13.24, [3, 5]	[4, 4], 12.61, [3, 5]	[3, 6], 13.83, [3, 5]	[4, 6], 12.39, [3, 5]	[4, 3], 12.83, [5, 4]
[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]
[6, 7], 22.0, [5, 6]				

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]

7. iterace

Open:

[3, 4], 13.24, [3, 5]	[4, 4], 12.61, [3, 5]	[3, 6], 13.83, [3, 5]	[4, 3], 12.83, [5, 4]	[6, 4], 18.0, [5, 4]
[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]	[6, 7], 22.0, [5, 6]
[3, 7], 21.71, [4, 6]				

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]				

8. iterace

Open:

[3, 4], 13.24, [3, 5]	[3, 6], 13.83, [3, 5]	[4, 3], 12.83, [5, 4]	[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]
[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]	[6, 7], 22.0, [5, 6]	[3, 7], 21.71, [4, 6]

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]	[4, 4], 12.61, [3, 5]			

9. iterace

Open:

[3, 4], 13.24, [3, 5]	[3, 6], 13.83, [3, 5]	[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]
[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]	[6, 7], 22.0, [5, 6]	[3, 7], 21.71, [4, 6]	[3, 2], 21.16, [4, 3]
[4, 2], 15.24, [4, 3]	[5, 2], 15.41, [4, 3]			

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]	[4, 4], 12.61, [3, 5]	[4, 3], 12.83, [5, 4]		

10. iterace

Open:

[3, 6], 13.83, [3, 5]	[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]
[5, 7], 16.08, [5, 6]	[6, 7], 22.0, [5, 6]	[3, 7], 21.71, [4, 6]	[3, 2], 21.16, [4, 3]	[4, 2], 15.24, [4, 3]
[5, 2], 15.41, [4, 3]				

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]	[4, 4], 12.61, [3, 5]	[4, 3], 12.83, [5, 4]	[3, 4], 13.24, [3, 5]	

11. iterace

Open:

[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]
[6, 7], 22.0, [5, 6]	[3, 7], 21.71, [4, 6]	[3, 2], 21.16, [4, 3]	[4, 2], 15.24, [4, 3]	[5, 2], 15.41, [4, 3]
[2, 7], 22.21, [3, 6]				

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]	[4, 4], 12.61, [3, 5]	[4, 3], 12.83, [5, 4]	[3, 4], 13.24, [3, 5]	[3, 6], 13.83, [3, 5]

12. iterace

Open:

[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]
[6, 7], 22.0, [5, 6]	[3, 7], 21.71, [4, 6]	[3, 2], 21.16, [4, 3]	[5, 2], 15.41, [4, 3]	[2, 7], 22.21, [3, 6]
[3, 1], 24.0, [4, 2]	[4, 1], 18.0, [4, 2]	[5, 1], 19.0, [4, 2]		

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]	[4, 4], 12.61, [3, 5]	[4, 3], 12.83, [5, 4]	[3, 4], 13.24, [3, 5]	[3, 6], 13.83, [3, 5]
[4, 2], 15.24, [4, 3]				

13. iterace

Open:

[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]
[6, 7], 22.0, [5, 6]	[3, 7], 21.71, [4, 6]	[3, 2], 21.16, [4, 3]	[2, 7], 22.21, [3, 6]	[3, 1], 24.0, [4, 2]
[4, 1], 18.0, [4, 2]	[5, 1], 19.0, [4, 2]	[6, 1], 16.0, [5, 2]	[6, 2], 21.0, [5, 2]	

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]	[4, 4], 12.61, [3, 5]	[4, 3], 12.83, [5, 4]	[3, 4], 13.24, [3, 5]	[3, 6], 13.83, [3, 5]
[4, 2], 15.24, [4, 3]	[5, 2], 15.41, [4, 3]			

14. iterace

Open:

[6, 4], 18.0, [5, 4]	[6, 5], 20.0, [5, 4]	[6, 6], 21.0, [5, 5]	[4, 7], 22.33, [5, 6]	[5, 7], 16.08, [5, 6]
[6, 7], 22.0, [5, 6]	[3, 7], 21.71, [4, 6]	[3, 2], 21.16, [4, 3]	[2, 7], 22.21, [3, 6]	[3, 1], 24.0, [4, 2]
[4, 1], 18.0, [4, 2]	[5, 1], 19.0, [4, 2]	[6, 2], 21.0, [5, 2]		

Closed:

[3, 5], 5.0, NULL	[4, 5], 8.47, [3, 5]	[5, 4], 10.16, [4, 5]	[5, 5], 11.12, [4, 5]	[5, 6], 12.10, [4, 5]
[4, 6], 12.39, [3, 5]	[4, 4], 12.61, [3, 5]	[4, 3], 12.83, [5, 4]	[3, 4], 13.24, [3, 5]	[3, 6], 13.83, [3, 5]
[4, 2], 15.24, [4, 3]	[5, 2], 15.41, [4, 3]			

15. iterace

Open:

Closed:

16. iterace

Open:

Closed:
